Introduction to Cybersecurity

Cyber-security incident response team

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Summary

- Before we start.
- What is a cyber security incident?
- How data breaches occur?
- What is incident response team?
- Reality check
- Do we need one?
- Incident response team roles
- What they are doing?
- Incident response model
- Tips for incident response team members
- Lecturer



Before we start. Do we/you have?

- Security in company management: CSO, CISO…
- Policy documents
- Security specialist or Ethical hacker
- Red team
- Blue team
- **Incident response teams**



What is a cyber security incident?

- When an organization's systems are compromised, you often hear the term "cyber security incident" – rather than "**breach**" or "**hack**"

- Unlike a breach, a cyber security incident doesn't necessarily mean information is compromised; it only means that information is threatened.



How Data Breaches Occur

1 Research



Attacker looks for weaknesses he can exploit

2 Stage Attack



SOCIAL ENGINEERING



Phishing email, spam with malware, phone call, dress like the night janitor, etc.

INFRASTRUCTURE WEAKNESS



FIREWALL WEBSITE WEB SERVER

SQL injection, vulnerability exploitation, session hijacking, etc.

Attacker may need to keep staging attacks until the desired information is obtained or the desired access to the network is achieved

3 Exfiltrate



DATABASE AND FILE SERVERS

Personally identifiable information (PII), credit card numbers, email addresses, other social details, etc.



HTTP-based path, FTP, email, etc.



Accessed data is exfiltrated back to attacker

Once the attacker maintains acess to the system, exfiltration can indefinitely proceed

What is incident response team?

- A computer security incident response team (CSIRT) is a concrete organizational entity that is assigned the responsibility for coordinating and supporting the response to a computer security event or incident.



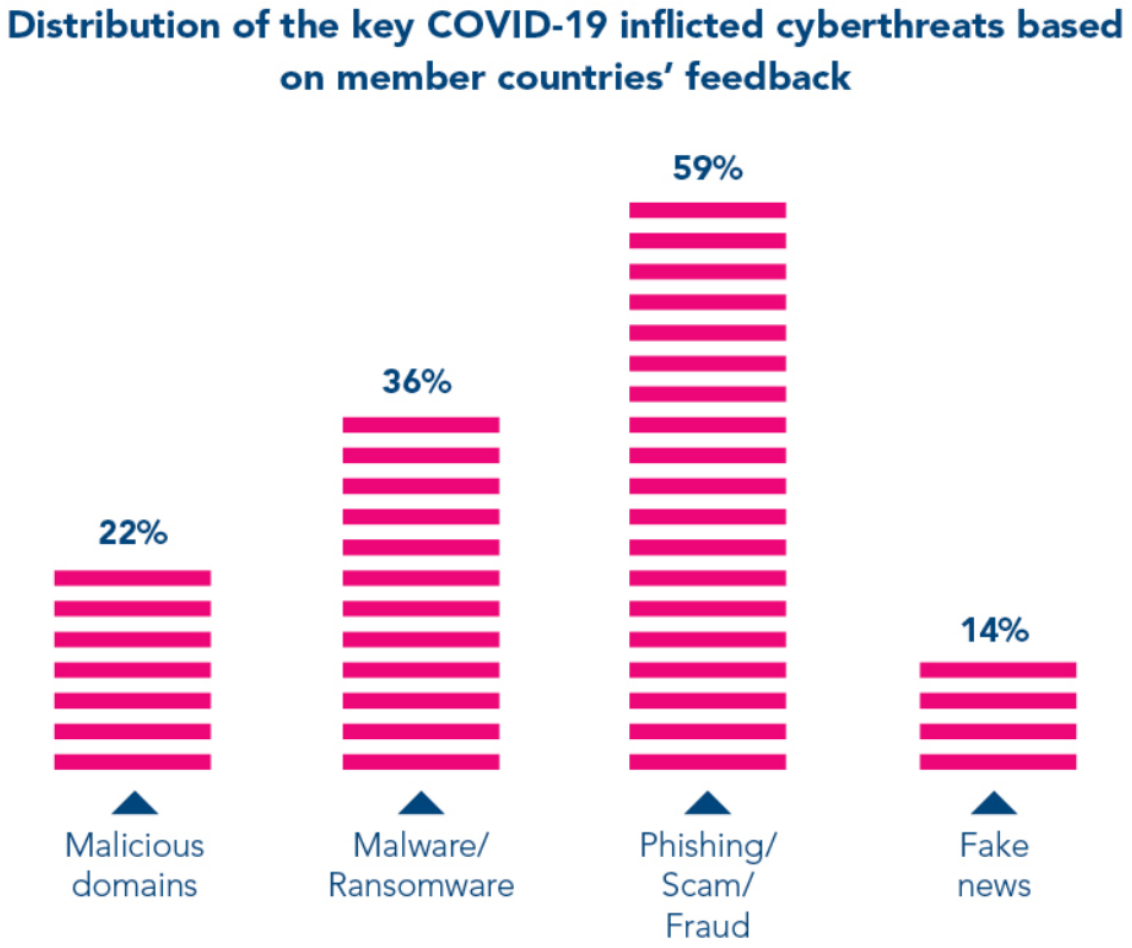
Reality check

"Cybercriminals are developing and boosting their attacks at an alarming pace, exploiting the fear and uncertainty caused by the unstable social and economic situation created by COVID-19."

Jürgen Stock, INTERPOL Secretary General

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Reality check



NEWS TECH

Report: Cyberattack on German hospital inadvertently causes patient's death

BY **NIC KER** 12:12 PM, 18 SEPTEMBER 2020 **2 COMMENTS**

Do we need one?

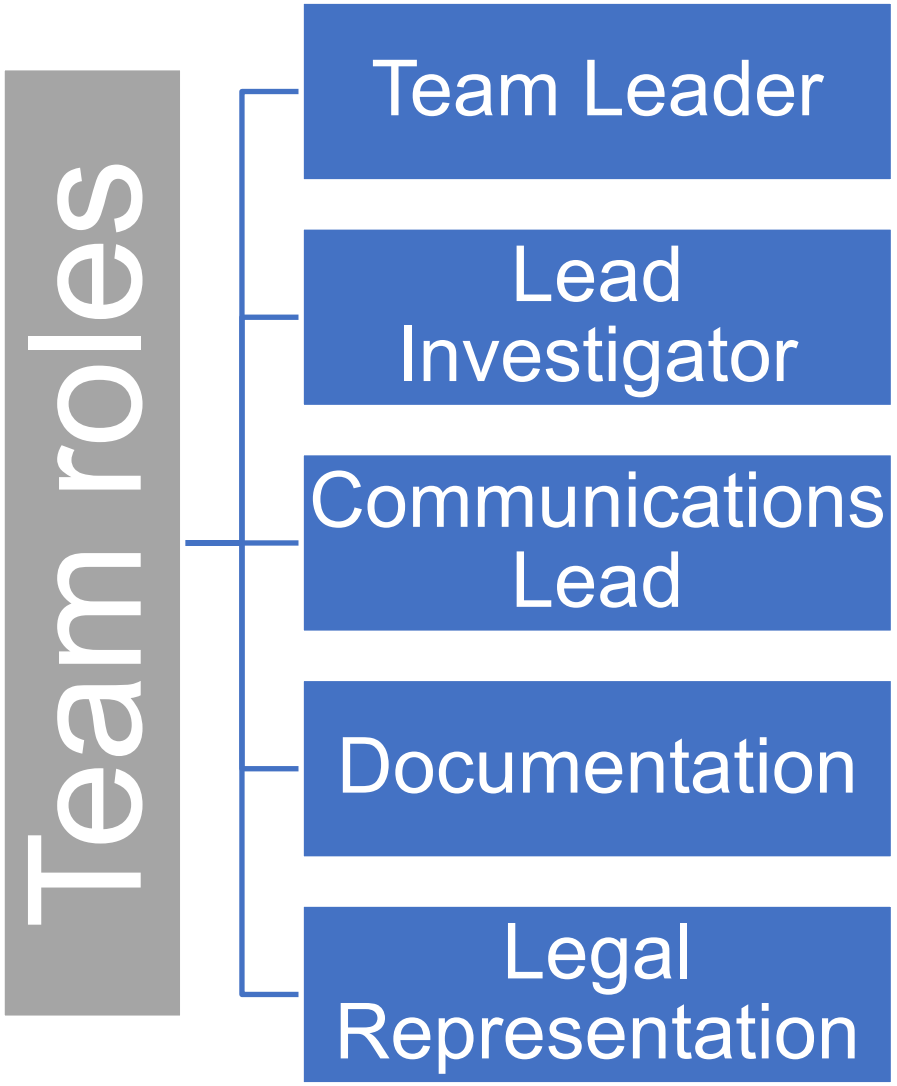


- *“Incident Response needs people, because successful Incident Response requires thinking.”* — Bruce Schneier, Schneier on Security

- Increased number of cyber-crime activities attempts in BiH
- **We need a plan and/or policy**

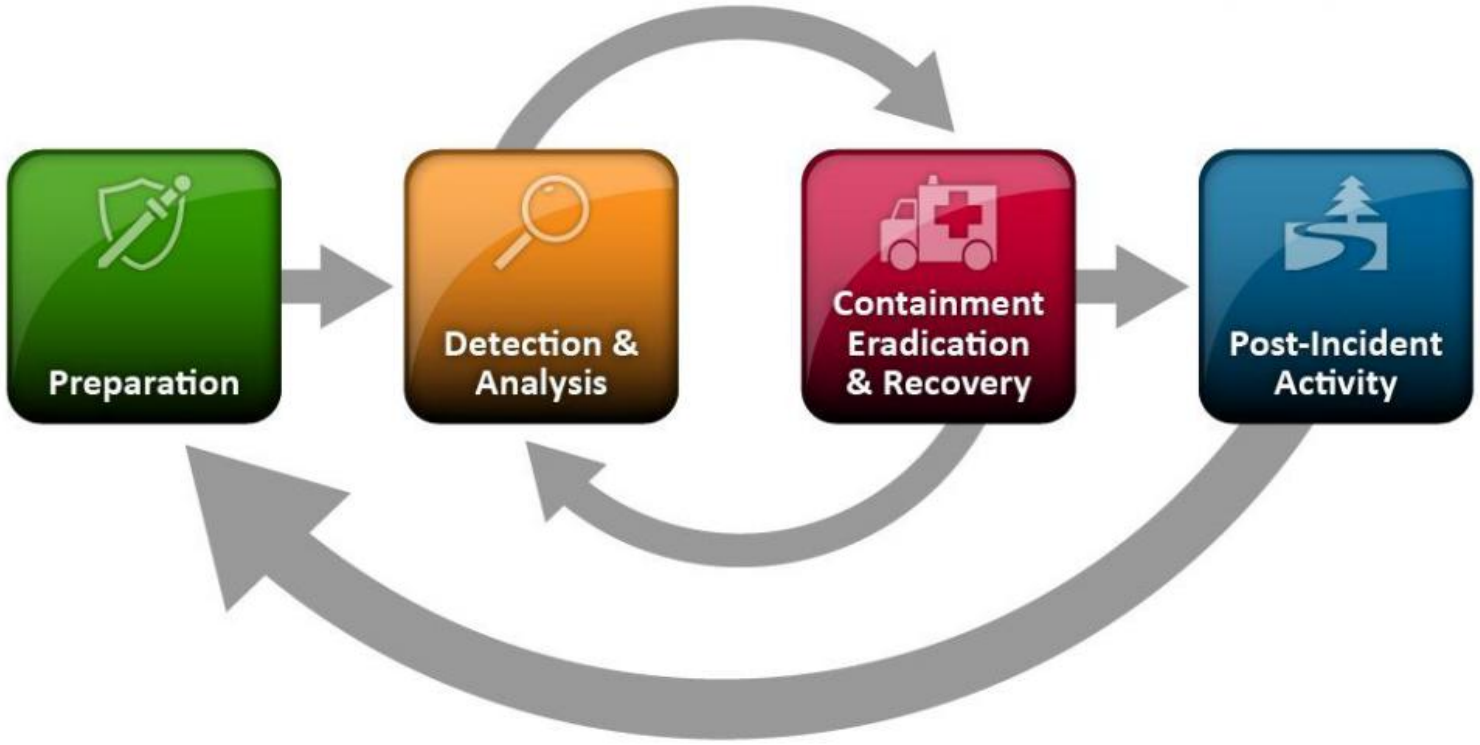


Incident response team roles

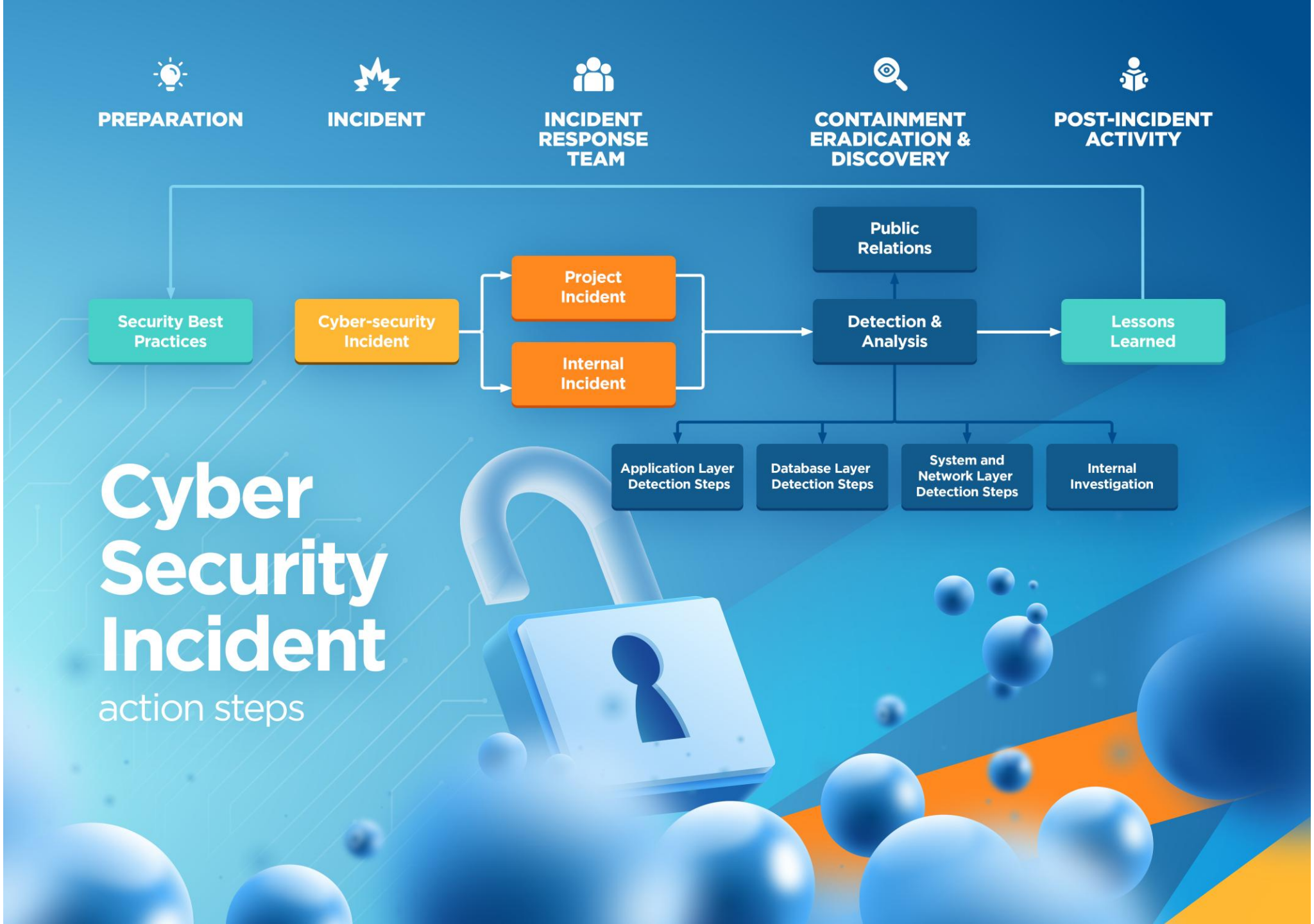


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What they are doing?






PREPARATION
INCIDENT
INCIDENT RESPONSE TEAM
CONTAINMENT ERADICATION & DISCOVERY
POST-INCIDENT ACTIVITY
Security Best Practices
Cyber-security Incident
Project Incident
Internal Incident
Public Relations
Detection & Analysis
Lessons Learned
Application Layer Detection Steps
Database Layer Detection Steps
System and Network Layer Detection Steps
Internal Investigation
Cyber Security Incident
action steps

Tips for incident response team members

- Look for the Common Denominators
- Look for the Common Exceptions
- Make Assertions, Not Assumptions
- Eliminate the Impossible
- Always Look for a Simpler Explanation
- Imagine things from the Attacker's Perspective what would you do in their position?



